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CANDIDATE – PLEASE NOTE!

You must sign below and return this booklet with the Answer Sheet. Failure to do so may result in disqualification.

Signature

TEST CODE **01234010**

MAY/JUNE 2011

FORM TP 2007104

**CARIBBEAN EXAMINATIONS COUNCIL
SECONDARY EDUCATION CERTIFICATE
EXAMINATION
MATHEMATICS**

Paper 01 – General Proficiency

90 minutes

18 MAY 2011 (p.m.)

READ THE FOLLOWING DIRECTIONS CAREFULLY

1. In addition to this test booklet, you should have an answer sheet.
2. Calculators and mathematical tables may NOT be used for this paper.
3. **A list of formulae is provided on page 2 of this booklet.**
4. This test consists of 60 items. You will have 90 minutes to answer them.
5. Each item in this test has four suggested answers, lettered (A), (B), (C), (D). Read each item you are about to answer, and decide which choice is best.
6. On your answer sheet, find the number which corresponds to your item and blacken the space having the same letter as the answer you have chosen. Look at the sample item below.

Sample Item

$$2a + 6a =$$

- (A) $8a$
- (B) $8a^2$
- (C) $12a$
- (D) $12a^2$

Sample Answer



The best answer to this item is “ $8a$ ”, so answer space (A) has been blackened.

7. If you want to change your answer, erase your old answer completely and fill in your new choice.
8. When you are told to begin, turn the page and work as quickly and as carefully as you can. If you cannot Answer an item, omit it and go on to the next one. You can return later to the item omitted. Your score will be the total number of correct answers.
9. You may do any rough work in the booklet.
10. Do not be concerned that the answer sheet provides spaces for more answers than there are items in this test.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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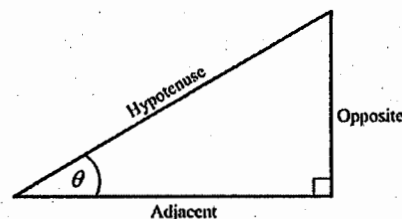
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LIST OF FORMULAE

Volume of a prism	$V = Ah$ where A is the area of a cross-section and h is the perpendicular length.
Volume of a cylinder	$V = \pi r^2 h$ where r is the radius of the base and h is the perpendicular height.
Volume of a right pyramid	$V = \frac{1}{3} Ah$ where A is the area of the base and h is the perpendicular height.
Circumference	$C = 2\pi r$ where r is the radius of the circle.
Arc length	$S = \frac{\theta}{360} \times 2\pi r$ where θ is the angle of the sector.
Area of a circle	$A = \pi r^2$ where r is the radius of the circle.
Area of a sector	$A = \frac{\theta}{360} \times \pi r^2$ where θ is the angle of the sector.
Area of Trapezium	$A = \frac{1}{2}(a+b)h$ where a and b are the lengths of the parallel sides and h is the perpendicular distance between the parallel sides.

Roots of quadratic equations If $ax^2 + bx + c = 0$, then $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Trigonometric ratios	$\sin \theta = \frac{\text{opposite side}}{\text{hypotenuse}}$
	$\cos \theta = \frac{\text{adjacent side}}{\text{hypotenuse}}$
	$\tan \theta = \frac{\text{opposite side}}{\text{adjacent side}}$

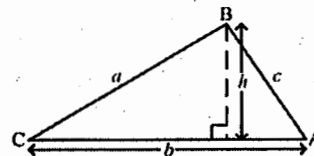


Area of triangle Area of $\Delta = \frac{1}{2}bh$ where b is the length of the base and h is the perpendicular height

$$\text{Area of } \Delta ABC = \frac{1}{2}ab \sin C$$

$$\text{Area of } \Delta ABC = \sqrt{s(s-a)(s-b)(s-c)}$$

$$\text{where } s = \frac{a+b+c}{2}$$



Sine rule $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

Cosine rule $a^2 = b^2 + c^2 - 2bc \cos A$

1. In scientific notation, 170.04 is written as

- (A) 0.17004×10^3
- (B) 1.7004×10^2
- (C) 17.004×10^1
- (D) 1.7004×10^{-1}

2. What percentage of 340 is 425

- (A) 80%
- (B) 85%
- (C) 125%
- (D) 152%

3. Ann and Betty shared a sum of money in the ratio 2:3 respectively. Ann received \$120. What was Betty's share?

- (A) \$ 72
- (B) \$ 80
- (C) \$180
- (D) \$300

4. $11.1 \div 0.01$ is equal to

- (A) 110
- (B) 111
- (C) 1100
- (D) 1110

5. If 60% of a number is 90, what is the number?

- (A) 30
- (B) 54
- (C) 150
- (D) 180

6. The EXACT value of $\frac{37.26 \div 1.8}{1000}$ is

- (A) 0.207
- (B) 0.0207
- (C) 20.7000
- (D) 20700

7. $-\left(\frac{1}{2}\right)^3$ is the same as

- (A) $-\frac{1}{8}$
- (B) $-\frac{1}{6}$
- (C) $\frac{1}{8}$
- (D) $\frac{1}{6}$

8. The H.C.F. of 12, 15 and 60 is

- (A) 1
- (B) 3
- (C) 12
- (D) 60

9. If $3n$ is an odd number, which of the following is an even number?

- (A) $3n-1$
- (B) $3n-2$
- (C) $3n+2$
- (D) $3n+2n$

10. What is the least number of plums that can be shared equally among 6, 9 or 12 children?

(A) 27
(B) 36
(C) 54
(D) 72

11. Which of the following sets is equivalent to $\{a, b, c, d\}$?

(A) $\{4\}$
(B) $\{a, b, c\}$
(C) $\{p, q, r, s\}$
(D) $\{1, 2, 3, 4, 5\}$

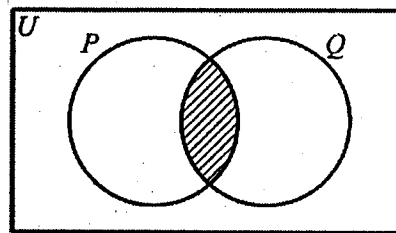
12. Which of the following sets is defined by $\{x \in \mathbb{Z} : -2 \leq x \leq 4\}$

(A) $\{1, 2, 3, 4\}$
(B) $\{0, 1, 2, 3, 4\}$
(C) $\{-1, 0, 1, 2, 3\}$
(D) $\{-2, -1, 0, 1, 2, 3, 4\}$

13. If $P = \{a, b\}$ then the number of subsets of P is

(A) 2
(B) 3
(C) 4
(D) 8

Item 14 refers to the Venn diagram below.



14. If $P = \{\text{Factors of 6}\}$ and $Q = \{\text{Factors of 4}\}$, then the shaded region represents

(A) $\{\}$
(B) $\{1, 2\}$
(C) $\{4, 6, 8, \dots\}$
(D) $\{12, 24, 36, \dots\}$

15. $3\frac{1}{4}\%$ of \$500 is

(A) \$ 1.62
(B) \$15.52
(C) \$16.00
(D) \$16.25

16. During a sale, a shop allows 20% discount off the marked price of clothing. What will a customer pay for a dress with a marked price of \$30?

(A) \$10
(B) \$20
(C) \$24
(D) \$30

17. A man bought a calf for \$200 and sold it for \$250. What was his gain as a percentage of the cost price?

(A) 5%
(B) 15%
(C) 20%
(D) 25%

18. How much does a customer pay for an article marked at \$50.00 before taxes if a sales tax of 6% is charged?
- (A) \$56.00
(B) \$53.00
(C) \$47.00
(D) \$44.00
19. At the end of any year, a car is worth 5% less than what it was worth at the beginning of the year. If a car was bought for \$10 000 in January 2009, its value in December 2009 was
- (A) \$9 000
(B) \$9 025
(C) \$9 500
(D) \$9 995
20. If the simple interest on \$800 for 3 years is \$54. What is the rate of interest per annum?
- (A) $\frac{4}{9}\%$
(B) $2\frac{1}{4}\%$
(C) 5%
(D) 44%
21. A man pays 60 cents for every 200 m³ of gas used, plus a fixed charge of \$13.25. How much does he pay when he uses 55 000 m³ of gas?
- (A) \$178.25
(B) \$175.25
(C) \$165.00
(D) \$151.25
22. If \$7000 is borrowed at the rate of 5% per annum for 3 years, the simple interest is
- (A) \$ 105
(B) \$ 210
(C) \$ 370
(D) \$1 050
23. $\frac{4}{5x} + \frac{2}{5x} =$
- (A) $\frac{6}{25x}$
(B) $\frac{8}{25x}$
(C) $\frac{6}{10x}$
(D) $\frac{6}{5x}$
24. $(x-2)(3x+4) =$
- (A) $3x^2 - 6x - 8$
(B) $3x^2 - 2x - 8$
(C) $3x^2 + 10x + 8$
(D) $3x^2 - 10x + 8$
25. If $5x - 26 = x + 50$ then the value of x is
- (A) -12
(B) -6
(C) 6
(D) 19

26. $3x^2 \times 2x^3 =$

- (A) $6x^5$
- (B) $6x^6$
- (C) $5x^6$
- (D) $72x^5$

27. $P = \frac{m^2}{2-m}$. When $m = -3$, the value of P is

- (A) 9
- (B) $\frac{9}{5}$
- (C) $-\frac{6}{5}$
- (D) -6

28. For $2x + 3 \geq 9$, the range of values of x is

- (A) $x \geq 3$
- (B) $x > 3$
- (C) $x \geq 6$
- (D) $x > 6$

29. John has x marbles and Max has twice as many. Max gives John 5 of his marbles. How many marbles does Max now have?

- (A) $x + 5$
- (B) $x - 5$
- (C) $2x + 5$
- (D) $2x - 5$

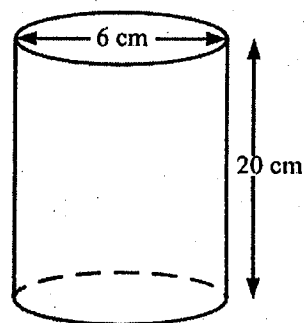
30. If $3 + \frac{2}{x} = 1$, then the value of x is

- (A) -1
- (B) $\frac{1}{5}$
- (C) 1
- (D) 5

31. The values of x and y which satisfy the equations $x + 2y = 27$ and $2x - y = 19$ are respectively

- (A) 15 and 10
- (B) 10 and 15
- (C) 7 and 13
- (D) 13 and 7

32. The diagram below shows a cylinder with diameter 6 cm and height 20 cm.



The volume in cm^3 , of the cylinder is

- (A) 180π
- (B) 240π
- (C) 360π
- (D) 720π

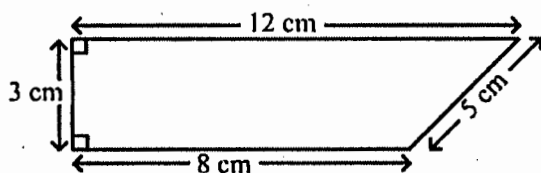
33. How many kilograms are there in one tonne?

- (A) 10
- (B) 100
- (C) 1 000
- (D) 10 000

34. The distance around the edge of a circular pond is 88 m. The radius, in metres is

- (A) 88π
- (B) 176π
- (C) $\frac{88}{\pi}$
- (D) $\frac{88}{2\pi}$

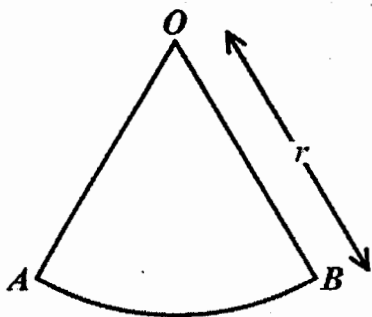
Item 35 refers to the quadrilateral below.



35. The area of the quadrilateral above is

- (A) 24 cm^2
- (B) 28 cm^2
- (C) 30 cm^2
- (D) 36 cm^2

Item 36 refers to the diagram below.



36. AOB is a sector of a circle such that angle $AOB = 60^\circ$ and OB is r units long. The area of AOB is

- (A) $\frac{1}{3}\pi r$
- (B) $\frac{1}{6}\pi r$
- (C) $\frac{1}{3}\pi r^2$
- (D) $\frac{1}{6}\pi r^2$

37. The area of a triangle is 30 cm^2 and its base is 10 cm . What is the perpendicular height, in cm , of the triangle?

- (A) 6
- (B) 12
- (C) 13
- (D) 17

38. The area of a rectangle is 53.6 cm^2 . If the length is multiplied by four and the width is halved, the area would then be

- (A) 26.8 cm^2
- (B) 53.6 cm^2
- (C) 107.2 cm^2
- (D) 214.4 cm^2

Item 39 refers to the table below showing the frequency of scores obtained by students in a test.

Scores	2	3	5	6	8	11
Students	8	4	6	3	12	2

39. The modal score is

- (A) 8
- (B) 9
- (C) 10
- (D) 12

40. The perimeter of a square is 48 cm . What is the area in cm^2 ?

- (A) 36
- (B) 72
- (C) 108
- (D) 144

41. The mean of the following numbers is 15.

14, 10, 18, c , 21, 15, 14

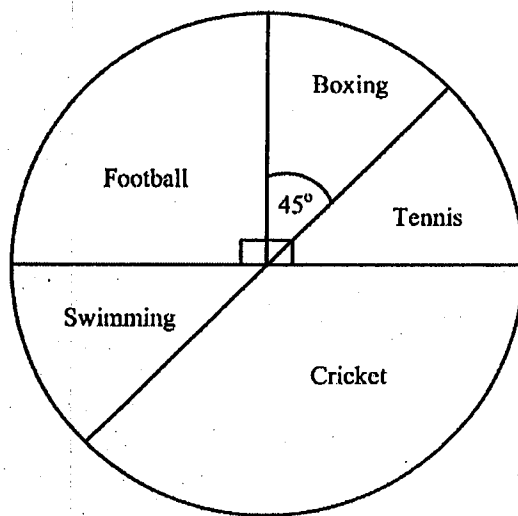
The value of c is.

- (A) 13
- (B) 14
- (C) 20
- (D) 91

42. A bag contains 2 red, 4 yellow and 6 blue balls. The probability of drawing a blue ball from the bag at random is

- (A) $\frac{1}{6}$
- (B) $\frac{1}{3}$
- (C) $\frac{1}{2}$
- (D) $\frac{6}{11}$

Items 43-45 refer to the diagram below which shows the sport chosen by 160 boys who participated in a games evening at their school



43. The number of boys who chose football is

- (A) 40
- (B) 90
- (C) 110
- (D) 150

44. The probability that a boy chosen at random participated in boxing is

- (A) $\frac{7}{8}$
- (B) $\frac{1}{2}$
- (C) $\frac{1}{4}$
- (D) $\frac{1}{8}$

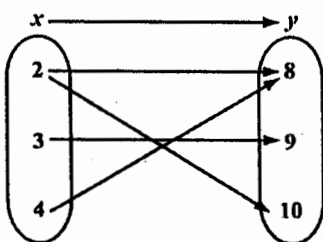
45. How many boys participated in cricket?

- (A) 54
- (B) 60
- (C) 110
- (D) 120

46. Which of the following represents the equation of a straight line?

- (A) $y = \frac{4}{x}$
 (B) $y = x^2 - 4$
 (C) $y = 2x + 3$
 (D) $y = x^2 + 2x - 5$

Item 47 refers to the arrow diagram below



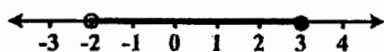
47. The arrow diagram above describes the relation

- (A) x is a factor of y
 (B) x is less than y
 (C) x is a multiple of y
 (D) x is greater than y

48. If $f(x) = 2x^2 - 1$ then $f(-3) =$

- (A) -32
 (B) -19
 (C) 17
 (D) 35

Item 49 refers to the diagram below.



49. The graph of the inequality in the diagram above is defined by

- (A) $-2 < x \leq 3$
 (B) $-2 \leq x < 3$
 (C) $-2 > x \geq 3$
 (D) $-2 \geq x > 3$

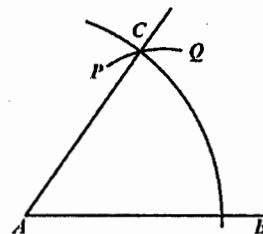
50. What is the gradient of the straight line $2y + 3x = -8$?

- (A) -3
 (B) $-\frac{3}{2}$
 (C) $\frac{2}{3}$
 (D) 3

51. Which of the following sets is represented by the relation $f: x \rightarrow x^2 + 3$?

- (A) $\{(0, 3), (1, 4), (2, 7), (3, 12)\}$
 (B) $\{(0, 3), (1, 5), (2, 7), (3, 9)\}$
 (C) $\{(0, 3), (1, 4), (2, 5), (3, 6)\}$
 (D) $\{(0, 3), (1, 1), (2, 4), (3, 9)\}$

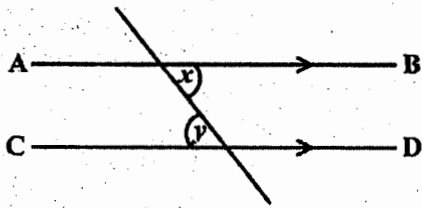
Item 52 refers to the diagram below of a construction. With centre A , an arc BC is drawn. With centre B , and the same radius, the arc PCQ is drawn.



52. What is the measure of $\angle BAC$?

- (A) 30°
 (B) 45°
 (C) 60°
 (D) 75°

Item 53 refers to the following diagram.

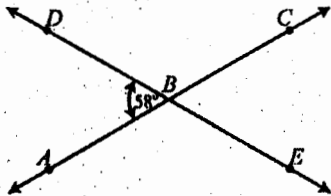


53. In the diagram, AB and CD are parallel. Which of the following BEST describes the relation between x and y ?

- (A) $x = y$
- (B) $x > y$
- (C) $x + y > 2x$
- (D) $x + y < 2x$

Item 54 refers to the diagram below.

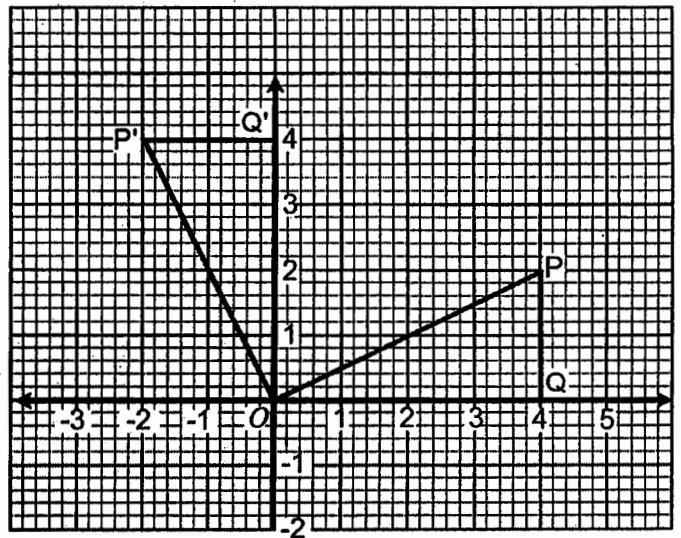
54. AC and DE are straight lines intersecting at B .
Angle $DBA = 58^\circ$



The measure of angle ABE is

- (A) 58°
- (B) 122°
- (C) 142°
- (D) 302°

Item 55 refers to the diagram below.

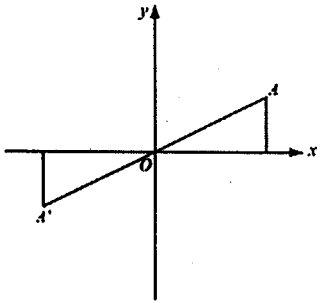


55. In the diagram above $\triangle OPQ$ is mapped onto $\triangle OP'Q'$. What type of transformation has taken place?

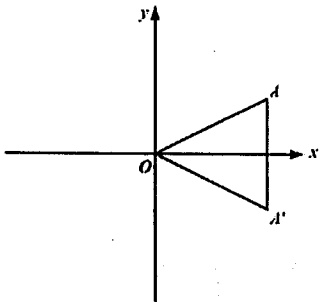
- (A) Reflection
- (B) Enlargement
- (C) Translation
- (D) Rotation

- 56 In each of the diagrams shown below, A' is the image of A . Which of the diagrams shows a reflection in the x -axis?

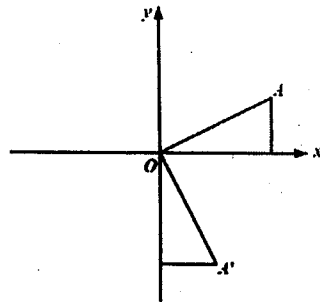
(A)



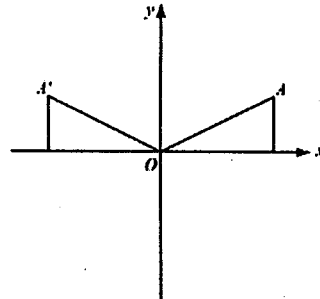
(B)



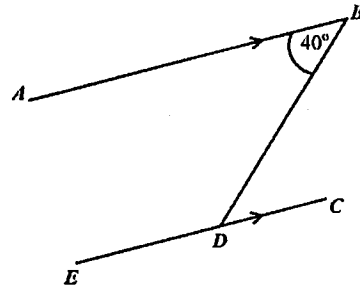
(C)



(D)



Item 57 refers to the diagram below.

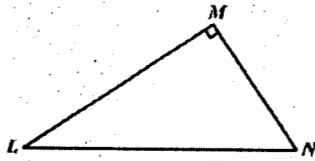


57. AB is parallel to EC . What is the measure of $\angle BDE$.

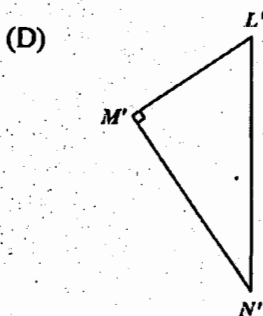
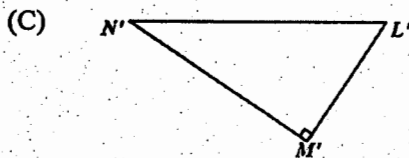
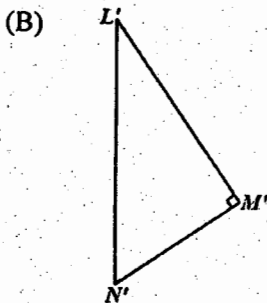
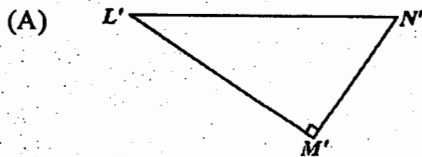
- (A) 40°
- (B) 50°
- (C) 140°
- (D) 180°

Item 58 refers to the triangle below.

58.

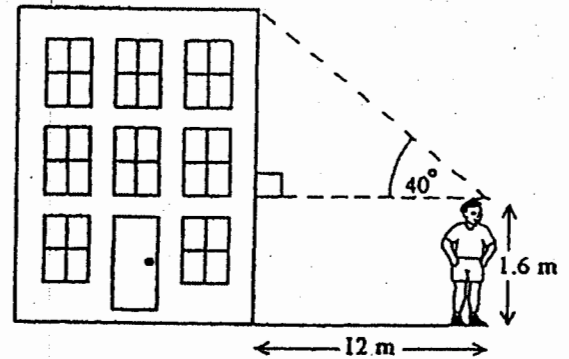


The triangle LMN is rotated in a clockwise direction about L through an angle of 90° . What is its image?



Item 59 refers to the diagram of the building below.

A boy stands 12 metres from the foot of the building and observes the angle of elevation of the top of the building.



59. The height of the building is approximately

- (A) $12 \tan 40^\circ$
- (B) $1.6 + 12 \sin 40^\circ$
- (C) $1.6 + 12 \cos 40^\circ$
- (D) $1.6 + 12 \tan 40^\circ$

60. A ladder 5 metres long is leaning against a vertical wall. The foot of the ladder is 3 m away from the wall on horizontal ground. How far up the wall does the ladder reach?

- (A) 4 m
- (B) 6 m
- (C) 8 m
- (D) 15 m

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.